YAARYATRA - Complete Feature Documentation

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Overview

YAARYATRA is a comprehensive ride-sharing and vehicle rental platform designed to connect travelers and vehicle owners, enabling cost-effective travel solutions through pooling services and flexible vehicle rentals. The platform serves both individual users and business owners (companies), providing a seamless experience for sharing rides, renting vehicles, and managing travel needs.

Key Value Propositions

- **For Individual Users**: Share travel costs through pooling, rent vehicles when needed, save money on transportation
- **For Company Owners**: Manage fleet, offer rental services, track bookings and earnings
- **For All Users**: Safe, verified, and reliable travel options with real-time tracking and communication

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Core Features

1. Multi-Language Support

Description: YAARYATRA supports multiple languages to cater to diverse user bases. The app currently supports English and Telugu (తెలుగు), with Hindi support planned for future releases.

Features:

- **Language Selection at Sign Up**: Users can select their preferred language before creating an account. The language selector appears on the Sign Up screen with a dropdown menu showing available languages (English/Telugu) with a globe icon indicator.
- **Language Persistence**: Selected language preference is saved to AsyncStorage for offline access and synced with the user's profile in the backend database.

- **Settings Integration**: Users can change their language preference anytime from Settings → Language, which opens a modal with language options. The selected language is marked with a checkmark.
- **Instant Language Switching**: The entire app UI updates immediately when a language is changed, without requiring an app restart. All UI strings, labels, buttons, and messages are dynamically translated.
- **Translation Infrastructure**: Uses `react-i18next` for internationalization with separate translation files (`en.json`, `te.json`, `hi.json`) for each supported language.

Technical Implementation:

- Language context provider manages global language state
- Translation keys used throughout the app for all user-facing text
- Language preference loaded on app startup and applied automatically
- Backend sync ensures language preference is maintained across devices

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Authentication & Registration

2. User Authentication System

Description: A comprehensive authentication system supporting multiple user types with secure login, registration, and password management.

Features:

- **Sign In**: Traditional username/password authentication with options for social login (Google, Phone)
- **Sign Up**: Multi-step registration process with user type selection (Individual/Company)
- **Forgot Password**: Three-step password reset flow:
    1. Enter phone/email to receive OTP
    2. Verify OTP sent via SMS/Email
    3. Set new password with confirmation
- **Remember Me**: Option to save login credentials for faster access
- **Session Management**: JWT-based authentication with access and refresh tokens
- **Security**: Password hashing using bcrypt, secure token storage, and session timeout

User Types:

- **Individual Users**: Personal accounts for pooling and rental services
- **Company Users**: Business accounts for fleet management and rental services

3. Individual User Registration

Description: Simplified two-step registration process for individual users, designed to minimize friction and get users started quickly.

Registration Flow:

1. **Step 1 - Phone Verification**:
    - User enters phone number with country code (+91 for India)
    - System sends OTP via SMS

    - User enters 4-digit OTP in separate input boxes
    - Resend OTP option with countdown timer (60 seconds)
    - Verification must be completed before proceeding

2. **Step 2 - Name Entry**:
    - User enters their full name
    - No email, password, date of birth, or gender required at registration
    - Documents are deferred until first service use (contextual document collection)
    - Registration completes and user is directed to main dashboard

Key Benefits:

- Minimal information required upfront
- Fast registration process (under 2 minutes)
- Documents collected only when needed (contextual verification)
- No email verification required initially

4. Company Registration

Description: Comprehensive registration process for business owners who want to offer rental services through the platform.

Registration Flow:

1. **Step 1 - Company Information**:
    - Company name (required)
    - Registration number (required)
    - Business type (dropdown selection)
    - Company address (required)
    - Contact number (required)
    - Email address (required)
2. **Step 2 - Company Documents**:
    - Company Registration Certificate
    - GST Certificate
    - Business License
    - Tax documents
3. **Step 3 - Account Credentials**:
    - Username (auto-generated from company name, can be edited)
    - Password (with strength requirements: 8+ characters, 1 uppercase, 1 number)
    - Confirm password

Verification Process:

- Company documents reviewed by admin
- Manual approval required
- Verification status tracked

- User notified upon approval/rejection

5. Document Verification System

Description: Contextual document collection system that requests documents only when users attempt to use specific services for the first time.

Document Requirements by Service:

- **Creating Pooling Offer**: Aadhar Card (Front & Back), User Photo, Driving License (Front & Back), Vehicle Number, Vehicle Photos (Front & Back), Insurance Papers
- **Creating Rental Offer**: Aadhar Card (Front & Back), User Photo, Driving License (Front & Back), Vehicle Number, Vehicle Photos (Front & Back), Insurance Papers
- **Taking Pooling Service**: Aadhar Card (Front & Back), User Photo
- **Taking Rental Service**: Aadhar Card (Front & Back), User Photo, Driving License (Front & Back)

Verification Flow:

1. User attempts to use a service (create offer or book service)
2. System checks if required documents are uploaded
3. If documents missing → Navigate to Document Verification Screen
4. User uploads required documents with camera or gallery
5. Documents submitted for admin review
6. User can proceed to service screen after submission
7. Admin reviews and approves/rejects documents
8. User notified of verification status

Features:

- Document upload with image picker (camera/gallery)
- Multiple document types supported
- Document status tracking (Pending, Verified, Rejected)
- Re-upload option for rejected documents
- Document expiry reminders (future feature)

Pooling Services

6. Create Pooling Offer

Description: Allows users to create ride-sharing offers where they can share their vehicle with other travelers going to the same destination, splitting travel costs.

Features:

- **Route Selection**: Pick from and to locations using interactive map or location picker
- **Date & Time Selection**: Choose departure date and time with date/time pickers
- **Vehicle Type Selection**: Choose between Car or Bike
- **Available Seats**: Set number of available seats (max based on vehicle capacity)
- **Dynamic Pricing**: System calculates price per person based on:
    - Route distance

    - Fuel costs
    - Vehicle type
    - Market rates
    - User can override with custom price
- **Additional Notes**: Optional field for special instructions (non-smoking, AC available, luggage space, etc.)
- **Pink Pooling**: Special pooling option exclusively for women and girls (gender verification required)

Offer Management:

- Offers listed in "My Offers" section
- Status tracking: Active, Pending, Expired, Completed, Cancelled
- Real-time seat availability updates
- View booking requests
- Accept/reject passenger requests
- Cancel offers with confirmation

Smart Features:

- Automatic offer expiration after trip date
- Seat availability auto-updates when bookings are made
- Route matching algorithm suggests similar routes
- Price suggestions based on market rates

7. Search & Book Pooling

Description: Comprehensive search and booking system for finding and joining pooling offers.

Search Features:

- **Location-Based Search**: Search by from/to locations
- **Date Filter**: Filter offers by specific date
- **Vehicle Type Filter**: Filter by Car or Bike
- **Advanced Filters**:
    - Price range (min/max)
    - Departure time (Morning, Afternoon, Evening)
    - Rating filter (4.5+, 4.0+, 3.5+)
    - Features (AC Available, Music System, Luggage Space)
    - Distance from user location
- **Sort Options**:
    - Price (Low to High / High to Low)
    - Rating (Highest first)
    - Distance (Nearest first)
    - Departure Time (Earliest first)

Search Results Display:

- List of matching pooling offers
- Driver information (name, rating, review count)
- Route details (from → to)

- Departure date and time
- Available seats count
- Price per person
- Vehicle type indicator
- Quick view details button

Booking Process:

1. User views pooling offer details
2. Sees driver profile, route, vehicle details, other passengers
3. Clicks "Join Pool" button
4. System checks document verification
5. Navigates to payment screen
6. User completes payment
7. Booking confirmed
8. Notification sent to driver and passenger

Pooling Details Screen:

- Complete driver information with profile link
- Route visualization (from → to)
- Trip date and time
- Vehicle details (type, number, model)
- Available seats count
- Price per person
- Additional notes from driver
- List of other confirmed passengers
- Join Pool button
- Message Driver button
- Share and favorite options

Rental Services

8. Create Rental Offer

Description: Allows vehicle owners (individuals and companies) to create rental offers, making their vehicles available for rent to other users.

Features:

- **Pickup Location**: Select pickup address with map integration
- **Date Selection**: Choose available date
- **Time Window**: Set available from/to times (e.g., 9:00 AM - 6:00 PM)
- **Vehicle Selection**: Choose from user's registered vehicles (dropdown)
- **Vehicle Type**: Car or Bike selection
- **Pricing**: Set price per hour with minimum rental hours requirement
- **Additional Notes**: Optional information about vehicle condition, features, etc.

Rental Management:

- View all rental offers in "My Offers"
- Track booking status (Available, Booked, Completed)
- See who booked the vehicle
- Manage availability calendar
- End rental early if needed
- View rental history and earnings

Company-Specific Features:

- Companies can manage multiple vehicles
- Bulk vehicle management
- Fleet overview dashboard
- Company earnings tracking
- Vehicle inventory management

9. Search & Book Rental

Description: Comprehensive search and booking system for finding and renting vehicles.

Search Features:

- **Location-Based Search**: Search rentals by pickup location
- **Date & Duration**: Select rental date and duration (hours)
- **Vehicle Type Filter**: Filter by Car or Bike
- **Advanced Filters**:
  - Price range per hour
  - Vehicle rating
  - Availability time slots
  - Distance from user location
  - Vehicle features

Search Results Display:

- Vehicle images (gallery view)
- Vehicle name and model
- Owner information and rating
- Price per hour
- Availability window
- Minimum rental hours
- Quick view details button

Booking Process:

1. User views rental details
2. Sees vehicle gallery, specifications, owner profile
3. Selects rental duration (hours)
4. Chooses time slot from available options
5. Views total price calculation
6. Clicks "Book Now"
7. System checks document verification

8. Navigates to payment screen
9. User completes payment
10. Booking confirmed
11. Notification sent to owner and renter

Rental Details Screen:

- Vehicle image gallery with swipe navigation
- Complete vehicle specifications:
  - Vehicle number
  - Type and seats
  - Fuel type and transmission
  - Year and model
- Owner profile with rating and reviews
- Pickup location with map
- Pricing breakdown (per hour, minimum hours, total)
- Duration selector
- Time slot selection
- Total amount calculation
- Book Now button
- Message Owner button

Availability Checking:

- Real-time availability verification
- Time slot conflict detection
- Automatic booking confirmation
- Calendar integration for availability

Payment System

10. Payment Integration

Description: Comprehensive payment system supporting multiple payment methods with secure transaction processing.

Supported Payment Methods:

- **UPI**: Unified Payments Interface (most popular in India)
- **Credit/Debit Cards**: Visa, Mastercard, RuPay
- **Digital Wallets**: Paytm, PhonePe, Google Pay, etc.
- **Net Banking**: Direct bank transfers

Payment Features:

- **Secure Payment Gateway**: Integration with Razorpay for payment processing
- **Payment Verification**: Automatic payment verification and confirmation
- **Payment Breakdown**: Transparent pricing showing:
  - Service charge

    - Platform fee (10% default, configurable)
    - Total amount
- **Payment History**: Complete transaction history for users
- **Refund Processing**: Automatic refund calculation and processing based on cancellation policy

Cancellation & Refund Policy:

- **Pooling Cancellations**:
    - Full refund if cancelled 24+ hours before trip
    - 50% refund if cancelled 12-24 hours before trip
    - No refund if cancelled less than 12 hours before trip
- **Rental Cancellations**:
    - Full refund if cancelled 48+ hours before rental
    - 50% refund if cancelled 24-48 hours before rental
    - No refund if cancelled less than 24 hours before rental

Payment Flow:

1. User selects payment method
2. Enters payment details (if required)
3. Confirms payment amount
4. Payment processed through gateway
5. Payment verification
6. Booking confirmed
7. Receipt generated
8. Notification sent

Security Features:

- PCI-DSS compliant payment processing
- Encrypted payment data transmission
- Secure token storage
- Payment webhook handling for status updates
- Fraud detection and prevention

Real-time Features

11. Trip Tracking

Description: Real-time location tracking system for active trips, enabling users to track their journey in real-time.

Features:

- **Live Map View**: Interactive map showing:
    - Driver's current location (real-time updates)
    - User's location
    - Destination marker
    - Route visualization

- **ETA Calculation**: Estimated time of arrival based on current location and traffic
- **Trip Statistics**:
  - Distance traveled
  - Duration elapsed
  - Remaining distance
  - Average speed
- **Communication**: Quick access to call and message driver
- **Safety Features**:
  - Emergency contact button
  - Share trip details with trusted contacts
  - Report issue functionality
- **Food Booking**: Quick access to book food along the route

Driver Trip Screen:

- Similar features for drivers
- View all passengers on the trip
- Start/end trip functionality
- Navigation integration
- Passenger management

Technical Implementation:

- WebSocket connection for real-time location updates
- GPS tracking with background location updates
- Map integration (Google Maps/OpenStreetMap)
- Location accuracy optimization
- Battery-efficient tracking

12. Real-time Notifications

Description: Comprehensive notification system keeping users informed about important events and updates.

Notification Types:

- **Booking Notifications**:
  - New booking requests
  - Booking confirmations
  - Booking cancellations
  - Booking reminders
- **Payment Notifications**:
  - Payment successful
  - Payment failed
  - Refund processed
- **Trip Notifications**:
  - Trip started
  - Trip completed
  - Driver location updates

- **Offer Notifications**:
  - New passenger joined (for drivers)
  - Offer expired
  - Offer cancelled
- **Rating Notifications**:
  - Rating request after trip completion
  - New rating received
- **System Notifications**:
  - Document verification status
  - Account updates
  - Platform announcements

Notification Features:

- **Push Notifications**: Real-time push notifications via Expo Notifications
- **In-App Notifications**: Notification center within the app
- **Email Notifications**: Optional email notifications for important events
- **SMS Notifications**: Optional SMS for critical updates
- **Notification Preferences**: User can customize which notifications to receive
- **Unread Count**: Badge showing unread notification count
- **Mark as Read**: Individual or bulk mark as read
- **Notification History**: View all past notifications

Communication Features

13. In-App Messaging (Chat)

Description: Real-time messaging system enabling communication between users (drivers/passengers, owners/renters).

Features:

- **Chat List**: View all active conversations
- **Real-time Messaging**: Instant message delivery via WebSocket
- **Message Types**:
  - Text messages
  - Location sharing
  - Image sharing (future)
  - Voice messages (future)
- **Chat Features**:
  - Typing indicators
  - Message read receipts
  - Timestamp display
  - Message status (sending, sent, delivered, read)
- **User Profiles**: Quick access to chat partner's profile from chat screen
- **Call Integration**: Direct call button from chat screen

Chat Contexts:

- Driver-Passenger communication (pooling)
- Owner-Renter communication (rental)
- Support chat with admin (future)

Technical Implementation:

- WebSocket connection for real-time messaging
- Message persistence in database
- Offline message queuing
- Message encryption for privacy

14. Rating & Review System

Description: Comprehensive rating and review system allowing users to rate their travel experiences.

Rating Features:

- **Overall Rating**: 1-5 star rating for overall experience
- **Detailed Ratings**: Rate specific aspects:
  - Punctuality (driver/owner)
  - Vehicle condition
  - Driving skills (for pooling)
  - Communication
  - Service quality
- **Written Reviews**: Optional text review describing the experience
- **Photo Uploads**: Optional photo uploads (future feature)
- **Response to Reviews**: Users can respond to reviews they receive

Rating Flow:

1. After trip completion, user receives rating request notification
2. User navigates to rating screen
3. Rates overall experience and specific aspects
4. Optionally writes a review
5. Submits rating
6. Rating displayed on user's profile
7. Average rating calculated and updated

Rating Display:

- User profiles show average rating and review count
- Individual ratings visible on profile
- Rating breakdown by category
- Recent reviews displayed

Profile & Settings

15. User Profile

Description: Comprehensive user profile management system.

Profile Features:

- **Personal Information**:
  - Name, phone, email
  - Date of birth, gender
  - Profile photo
- **Account Status**:
  - Verification status
  - Document status
  - Account status (Active/Suspended)
- **Statistics**:
  - Total trips completed
  - Average rating
  - Total earnings (for offerers)
  - Total spent (for service takers)
- **Vehicle Management**: View and manage registered vehicles
- **Document Management**: View uploaded documents and verification status
- **Activity Summary**: Recent transactions and activity

Profile Actions:

- Edit profile information
- Update profile photo
- View transaction history
- Manage vehicles
- View documents
- Access settings

16. Vehicle Management

Description: Complete vehicle registration and management system for users who own vehicles.

Add Vehicle Features:

- **Vehicle Information**:
  - Vehicle type (Car/Bike)
  - Make and model
  - Year of manufacture
  - Color
  - Vehicle number (registration number)
  - Seats capacity
  - Fuel type (Petrol, Diesel, Electric, CNG)
  - Transmission (Manual, Automatic)
- **Vehicle Photos**: Upload multiple photos:
  - Front view
  - Back view
  - Side view
  - Interior view
- **Documents**:

    - Registration Certificate
    - Insurance Papers
    - Pollution Certificate
    - Taxi Service Papers (for companies)
- **Company Vehicles**: Companies can add multiple vehicles with company association

Vehicle Management:

- View all registered vehicles
- Edit vehicle information
- Update vehicle photos
- View vehicle documents
- Delete vehicles
- Vehicle status tracking (Available, Booked, Maintenance)

Company Vehicle Management:

- Fleet overview dashboard
- Vehicle inventory management
- Bulk vehicle operations
- Vehicle availability calendar
- Vehicle utilization statistics

17. Settings

Description: Comprehensive settings system for app customization and account management.

Settings Categories:

- **Account Settings**:
    - Edit profile
    - Change password
    - Privacy settings
- **Notification Settings**:
    - Booking updates (on/off)
    - Messages (on/off)
    - Promotions (on/off)
- **App Preferences**:
    - Language selection (English/Telugu/Hindi)
    - Theme selection (Light/Dark - future)
- **Payment Settings**:
    - Payment methods management
    - Transaction history
- **Support**:
    - Help center
    - Contact us
    - Report issue
    - Feedback submission
- **About**:

    - Terms & Conditions
    - Privacy Policy
    - App version

Language Settings:

- Select from available languages
- Instant app-wide language change
- Language preference saved and synced

Admin Module

18. Admin Dashboard

Description: Comprehensive admin panel for platform management, monitoring, and analytics.

Dashboard Features:

- **User Analytics Panel**:
    - Present user count (real-time active users)
    - Total user count (cumulative registered users)
    - User breakdown (Individual/Company)
    - Verification status breakdown
    - User growth trends
- **Earnings Overview Panel**:
    - Today's earnings with comparison to previous day
    - Total earnings (lifetime)
    - Service-wise breakdown:
        - Car Pooling earnings
        - Bike Pooling earnings
        - Car Rentals earnings
        - Bike Rentals earnings
        - Corporate earnings
- **Feedback/Inquiry Panel**:
    - Recent feedback items
    - Status indicators (Pending, Acknowledged, Resolved)
    - Priority levels
    - Quick action buttons
- **Transactions Table**:
    - Complete transaction history
    - User name, service type, date, revenue
    - Filtering and pagination
    - Export options

19. Pooling Management

Description: Admin tools for managing pooling offers across the platform.

Features:

- **Offer Management**:
  - View all pooling offers
  - Filter by status (All, Active, Pending, Expired, Suspended, Flagged)
  - Statistics summary (total offers, active, pending, etc.)
- **Offer Actions**:
  - Approve offers
  - Suspend offers
  - Flag suspicious offers
  - View offer details
  - Contact driver
- **Bulk Operations**:
  - Select multiple offers
  - Bulk approve
  - Bulk suspend
  - Bulk export
- **Offer Details View**:
  - Complete offer information
  - Driver profile and rating
  - Passenger list
  - Interaction metrics (views, bookings)
  - Admin action buttons

20. Rental Management

Description: Admin tools for managing rental offers across the platform.

Features:

- **Rental Management**:
  - View all rental offers
  - Filter by status and type (Individual/Company)
  - Statistics summary
- **Rental Actions**:
  - Approve rentals
  - Suspend rentals
  - Flag suspicious rentals
  - View rental details
  - Contact owner
- **Bulk Operations**: Similar to pooling management
- **Rental Details View**:
  - Complete rental information
  - Owner profile and rating
  - Vehicle details
  - Booking statistics
  - Revenue generated

21. User Management

Description: Comprehensive user management system for admin oversight.

Features:

- **User List**:
  - View all users with filters (All, Individual, Company, Verified, Pending, Suspended)
  - User statistics summary
  - User details table with:
    - User ID, Name, Type, Status, Joined Date
- **User Actions**:
  - Verify users
  - Suspend users
  - Unverify users
  - Contact users
  - View user details
- **User Details View**:
  - Complete user profile
  - Account status
  - Activity summary
  - Vehicle information
  - Recent transactions
  - Document status
- **Bulk Operations**:
  - Bulk verify
  - Bulk suspend
  - Bulk export

22. Feedback Management

Description: System for managing user feedback, complaints, and suggestions.

Features:

- **Feedback List**:
  - View all feedback with filters (All, Pending, Acknowledged, Resolved, Archived)
  - Feedback statistics
  - Type indicators (Payment Issue, Feature Suggestion, Complaint)
  - Priority levels (Critical, High, Medium, Low)
- **Feedback Details**:
  - User information
  - Feedback details (type, category, subject, description)
  - Conversation history
  - Related transaction link
  - Admin response functionality
- **Feedback Actions**:
  - Acknowledge feedback

    - Mark as resolved
    - Assign to support team
    - Escalate
    - Archive
    - Send response to user

23. Analytics & Reports

Description: Comprehensive analytics and reporting system for business insights.

Features:

- **Key Metrics**:
    - New users (with growth percentage)
    - Revenue (with growth percentage)
    - Total trips
- **Charts & Visualizations**:
    - Revenue trends (line/bar chart)
    - User growth chart
    - Service distribution (pie chart)
- **Reports**:
    - Top performing routes
    - Top earning users
    - Service-wise breakdown
- **Custom Reports**:
    - Select date range
    - Choose metrics
    - Apply filters
    - Export formats (CSV, PDF, Excel)
    - Schedule recurring reports

24. Rides History (Admin)

Description: Complete transaction history management for admin oversight.

Features:

- **Transaction List**:
    - All completed transactions (Pooling + Rentals)
    - Filter by user, location, date range, revenue, service type, status
    - Transaction summary statistics
- **Transaction Details**:
    - Complete booking information
    - User details with profile link
    - Payment breakdown
    - Review and rating
    - Document links
- **Export Options**:
    - Export to CSV

    - Export to PDF
    - Generate custom reports

25. Admin Settings

Description: System configuration and admin account management.

Features:

- **Platform Configuration**:
    - Platform fee percentage (default: 10%)
    - Minimum booking amount
    - Maximum booking amount
- **Verification Settings**:
    - Auto-approve after hours (default: 24)
    - Require manual approval toggle
- **Notification Settings**:
    - Email notifications (on/off)
    - SMS notifications (on/off)
- **Admin Account**:
    - Change password
    - Two-factor authentication
    - Activity log access

Safety Features

32. Pink Pooling (Women-Only Pooling)

Description: A specialized pooling service exclusively designed for women and girls, providing a safe and secure travel option. Pink Pooling ensures that only female drivers and passengers can participate, creating a comfortable and secure environment for women travelers.

Key Features:

- **Gender Verification System**:

    - System automatically verifies user gender from profile before allowing Pink Pooling offer creation
    - Backend validation ensures only users with gender marked as "Female" can create Pink Pooling offers
    - If a non-female user attempts to create a Pink Pooling offer, the system returns a 403 error with message "Pink Pooling is only available for women and girls"
    - Gender information is stored in user profile and verified at offer creation time

- **Exclusive Access Control**:

    - Only female users can create Pink Pooling offers
    - Only female users can search and join Pink Pooling offers
    - Search results automatically filter to show only female drivers when Pink Pooling mode is active
    - Backend query filters offers by `driverGender: 'Female'` when `pinkOnly` parameter is true

- **Pink Pooling Mode**:
  - Special theme activation: When users enter Pink Pooling, the app switches to a pink-themed UI
  - Pink gradient colors (#FFDEE7 to #FF87A8) applied throughout the interface
  - Visual indicators: Pink badges and banners show "Pink Pooling Mode" is active
  - Splash Screen: Dedicated animated splash screen when entering Pink Pooling mode
  - Easy Exit: Users can exit Pink Pooling mode to return to normal pooling
- **User Interface**:
  - **Main Dashboard**: Pink Pooling button visible only to female users with heart icon
  - **Splash Screen**: Animated Lottie splash screen (3 seconds) before entering Pink Pooling mode
  - **Search Screen**:
    - Banner showing "Showing only female drivers - Pink Pooling Mode"
    - Pink badge on each offer card indicating Pink Pooling
    - All offers filtered to show only female drivers
  - **Theme Integration**: Complete UI theme change with pink color scheme
- **Complete Feature Parity**:
  - All standard pooling features available:
    - Route selection (from/to locations)
    - Date and time selection
    - Vehicle type selection (Car/Bike)
    - Available seats management
    - Dynamic pricing
    - Booking system
    - Payment integration
    - Trip tracking
    - Rating and reviews
- **Safety & Security**:
  - Verified female drivers only
  - Verified female passengers only
  - Same document verification requirements
  - Same rating and review system
  - Same reporting and blocking features
- **Technical Implementation**:
  - Backend flag: `isPinkPooling: boolean` in PoolingOffer model
  - Driver gender stored: `driverGender: 'Male' | 'Female' | 'Other'` in offer
  - Search filter: `pinkOnly: boolean` parameter in search API
  - Theme context: `isPinkMode` state in ThemeContext
  - Persistent theme: Pink mode preference saved to AsyncStorage

User Flow:

1. Female user sees Pink Pooling button on main dashboard

2. User taps Pink Pooling button
3. Pink Pooling splash screen appears with animation
4. App switches to Pink Pooling mode (pink theme activated)
5. User navigates to main dashboard in Pink Pooling mode
6. User can create or search Pink Pooling offers
7. All offers shown are from female drivers only
8. User can exit Pink Pooling mode anytime to return to normal pooling

Benefits:

- Enhanced safety for women travelers
- Comfortable environment for female users
- Peace of mind for parents and guardians
- Verified female-only community
- Same cost-saving benefits as regular pooling

33. SOS & Emergency Features

Description: Comprehensive safety and emergency response system designed to keep users safe during trips. The SOS feature provides quick access to emergency contacts and help during critical situations.

Emergency Contact Feature:

- **Quick Access Button**:
  - Prominently displayed on Trip Tracking screen during active trips
  - Always accessible, even when app is in background
  - One-tap access to emergency contacts
  - Located in emergency container section alongside other safety features
- **Emergency Contact Management**:
  - Users can add trusted emergency contacts in their profile
  - Multiple emergency contacts supported
  - Contact information includes name, phone number, and relationship
  - Quick dial functionality to call emergency contacts directly
- **Emergency Actions**:
  - **Call Emergency Contact**: Direct phone call to saved emergency contacts
  - **Share Trip Details**: Automatically share current trip location and details with emergency contacts via SMS/WhatsApp
  - **Send Location**: Share real-time GPS location with emergency contacts
  - **Trip Information Sharing**: Includes:
    - Current location (GPS coordinates)
    - Driver information (name, phone, vehicle details)
    - Route information (from/to locations)
    - Trip status and ETA
    - Booking ID for reference

Report Issue Feature:

- **Issue Reporting**:
  - Quick access button on Trip Tracking screen
  - Report various types of issues:
    - Safety concerns
    - Vehicle problems
    - Driver behavior issues
    - Route deviations
    - Payment disputes
    - Other concerns
  - Detailed issue description with optional photo attachments
  - Automatic trip context included (booking ID, driver info, location)
- **Issue Tracking**:
  - Reported issues tracked in user's history
  - Status updates (Pending, Under Review, Resolved)
  - Admin response and resolution tracking
  - Follow-up communication options

Safety Features During Trips:

- **Real-time Location Sharing**:
  - Continuous location tracking during active trips
  - Option to share live location with trusted contacts
  - Location updates sent automatically at regular intervals
  - Emergency contacts receive alerts if user doesn't respond
- **Trip Details Sharing**:
  - Share complete trip information before trip starts
  - Includes driver details, vehicle information, route, and timing
  - Recipients can track trip progress
  - Automatic notifications to emergency contacts when trip starts
- **Quick Communication**:
  - Direct call button to driver/owner
  - In-app messaging for non-emergency communication
  - Emergency contact quick dial
  - Support hotline access

Emergency Response System:

- **Automated Alerts**:
  - Panic button functionality (future enhancement)
  - Automatic location sharing in emergency situations

    - SMS alerts to emergency contacts
    - Push notifications to emergency contacts

- **Support Integration**:
    - Direct connection to platform support team
    - 24/7 emergency support hotline
    - Priority handling for emergency reports
    - Coordination with local authorities if needed

Privacy & Control:

- **User Control**:
    - Users choose which contacts to add as emergency contacts
    - Control when to share location and trip details
    - Option to disable automatic sharing
    - Privacy settings for location sharing
- **Data Protection**:
    - Emergency contacts stored securely
    - Location data encrypted
    - Shared information limited to necessary details
    - GDPR compliant data handling

Accessibility:

- **Easy Access**:
    - Large, easily accessible buttons
    - Works even with phone locked (future enhancement)
    - Voice command support (future enhancement)
    - Quick gesture shortcuts (future enhancement)

Integration with Trip Tracking:

- Emergency features available throughout active trip
- Location tracking continues during emergencies
- Real-time updates to emergency contacts
- Seamless integration with trip tracking map

Future Enhancements:

- Panic button with automatic alert system
- Integration with local emergency services (police, ambulance)
- Automatic incident detection using AI
- Emergency ride cancellation and alternative arrangements
- Integration with wearable devices for hands-free emergency access

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Additional Features

26. Food Booking Along Route

Description: Innovative feature allowing users to discover and book food from restaurants along their travel route.

Features:

- **Route-Based Discovery**: Automatically shows food shops between source and destination
- **Time-Based Filtering**:
  - Current (automatically filters based on current time)
  - Tiffin (6 AM - 11 AM)
  - Lunch (11 AM - 4 PM)
  - Dinner (4 PM - 11 PM)
- **Shop Information**:
  - Shop name and rating
  - Category badge (Tiffin/Lunch/Dinner)
  - Address and location
  - Operating hours
  - Direct call button
- **Location Grouping**: Shops grouped by intermediate locations along the route
- **Empty States**: Helpful messages when no shops found

Use Cases:

- Book tiffin for morning trips
- Find lunch spots during long journeys
- Discover dinner options for evening travel
- Call restaurants directly to place orders

27. Advanced Search & Filtering

Description: Comprehensive search and filtering system for finding the perfect travel options.

Filter Options:

- **Price Range**: Slider for minimum and maximum price
- **Vehicle Type**: Checkboxes for Car and Bike
- **Rating**: Filter by minimum rating (4.5+, 4.0+, 3.5+)
- **Departure Time**: Morning, Afternoon, Evening time slots
- **Features**: AC Available, Music System, Luggage Space
- **Sort Options**:
  - Price (Low to High / High to Low)
  - Rating (Highest first)
  - Distance (Nearest first)
  - Departure Time (Earliest first)

Search Features:

- Location-based search with autocomplete
- Recent searches saved

- Popular locations quick access
- Map integration for location selection

28. Booking History

Description: Complete booking history management for users to track all their travel activities.

Features:

- **Filter Tabs**: All, Upcoming, Past, Cancelled
- **Booking Cards**:
  - Service type (Pooling/Rental)
  - Route or vehicle details
  - Date and time
  - Status badge (Confirmed, Completed, Cancelled)
  - Amount paid
- **Booking Details**:
  - Complete booking information
  - Driver/Owner details
  - Vehicle information
  - Payment details
  - Trip tracking (for active trips)
  - Rating option (for completed trips)
- **Actions**:
  - View details
  - Track trip (if active)
  - Cancel booking (if allowed)
  - Rate trip (if completed)
  - Contact driver/owner

Company History:

- Separate history view for companies
- Earnings tracking
- Booking statistics
- Filter by date range and status

29. Help & Support

Description: Comprehensive help and support system for user assistance.

Features:

- **Search Help Topics**: Search functionality for finding help articles
- **Popular Topics**:
  - How to create a pooling offer
  - How to book a rental vehicle
  - Payment issues
  - Cancellation policy

- **Contact Support**:
  - Live chat (future)
  - Call support (phone number)
  - Email support (email address)
- **Additional Resources**:
  - FAQs
  - Report a bug
  - Feedback submission

30. Feedback System

Description: User feedback collection system for continuous improvement.

Features:

- **Feedback Types**:
  - Complaint
  - Suggestion
  - Compliment
- **Feedback Submission**:
  - Subject line (required)
  - Detailed message (required)
  - Optional attachments (future)
- **Feedback Management**:
  - Admin can view and respond
  - Status tracking (Pending, Acknowledged, Resolved)
  - User notified of responses

31. Location Picker

Description: Advanced location selection system with multiple input methods.

Features:

- **Map View**: Interactive map with draggable marker
- **Search**: Text search for locations with autocomplete
- **Current Location**: Quick access to use current GPS location
- **Recent Locations**: Saved recent location selections
- **Popular Locations**: Quick access to frequently used locations
- **Location Details**: Shows full address, city, state, coordinates

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Technical Stack

Mobile App (React Native)

- **Framework**: React Native with Expo
- **Navigation**: React Navigation (Stack & Tab Navigators)
- **State Management**: React Context API
- **Internationalization**: react-i18next

- **UI Components**: Custom components with React Native
- **Maps**: Expo Location, Map integration
- **Notifications**: Expo Notifications
- **Storage**: AsyncStorage, SecureStore
- **WebSocket**: Custom WebSocket service for real-time features
- **HTTP Client**: Axios for API calls

Backend (Node.js)

- **Framework**: Fastify
- **Database**: MongoDB with Mongoose
- **Authentication**: JWT (JSON Web Tokens)
- **File Storage**: Cloudinary for document and image storage
- **Payment Gateway**: Razorpay integration
- **Email Service**: Nodemailer
- **SMS Service**: Custom SMS service integration
- **WebSocket**: Fastify WebSocket for real-time features
- **Validation**: Zod for request validation
- **Logging**: Winston for application logging
- **Security**: Helmet, CORS, Rate limiting

Key Technologies

- **Real-time Communication**: WebSocket for chat and location tracking
- **Payment Processing**: Razorpay payment gateway
- **Document Management**: Cloudinary for document storage and processing
- **Location Services**: GPS tracking, geocoding, route calculation
- **Push Notifications**: Expo Push Notifications
- **Image Processing**: Expo Image Picker, Cloudinary transformations

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Security Features

Authentication & Authorization

- JWT-based authentication with refresh tokens
- Password hashing using bcrypt
- Secure token storage
- Session management
- Role-based access control (Individual, Company, Admin)

Data Protection

- Encrypted data transmission (HTTPS)
- Secure document storage
- PII (Personally Identifiable Information) protection
- GDPR compliance considerations
- Secure payment processing (PCI-DSS compliant)

Platform Security

- Rate limiting on API endpoints
- Input validation and sanitization
- SQL injection prevention (MongoDB)
- XSS (Cross-Site Scripting) protection
- CSRF (Cross-Site Request Forgery) protection

Performance Optimizations

Mobile App

- Image optimization and lazy loading
- Code splitting and lazy loading
- Efficient state management
- Cached API responses
- Optimized re-renders

Backend

- Database query optimization
- Caching strategies
- Pagination for large datasets
- Background job processing
- Efficient file upload handling

Future Enhancements

Phase 2 Features

- AI-powered route optimization
- Preference learning
- Smart pricing suggestions
- Social features (user groups, trip sharing)
- Loyalty program (points, rewards, discounts)
- Advanced analytics (personal insights, cost savings, environmental impact)

Phase 3 Features

- Calendar integration
- Advanced map integration
- Scheduled recurring trips
- Group bookings
- Corporate accounts
- IoT integration (smart vehicle tracking, automated check-in)

Conclusion

YAARYATRA is a comprehensive ride-sharing and vehicle rental platform with extensive features designed to provide a seamless, secure, and user-friendly experience. The platform serves individual users, company owners, and administrators with specialized features for each user type. With real-time tracking, secure payments, comprehensive communication tools, and robust admin management, YAARYATRA aims to revolutionize how people share rides and rent vehicles.

Document Version: 1.0
Last Updated: 2024
Total Features: 33+
Supported Languages: English, Telugu (Hindi coming soon)
Platform: iOS, Android, Web (future)